Appendix
VeriSAGE: Verified Agentic Service Composition

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This appendix accompanies the main paper and collects theory detail deferred for space: proofs of the main-paper theorems, a relative-completeness corollary, and the tighter and correlation-robust SLA bounds that extend Theorem 3. All cross-references resolve to the main paper.

1 Proofs

Proof (Proof of Theorem 1 (Execution soundness)). Algorithm 1 returns a plan only when *ok*, set iff all of C1 to C5 succeed; it suffices that each check reports success only when its predicate holds. **C1**: subtyping $\preceq$ is decidable and each edge is accepted only when $R(q_{i-1}) \preceq \mathsf{in}(s_i)$, so admitted plans call only existing operations with well-typed data. **C2** accepts only when an SMT solver proves $\mathsf{pre}(s_i)$ at q_{i-1} and $q_k \models \mathsf{post}_G$, and SMT is sound [7]. **C3** accepts iff each $\varphi \in \Phi$'s NFA for $\neg\varphi$ rejects $\pi(\mathcal{C})$; since Φ is the trace-closed safety fragment and C3 rejects any I_Φ-dependent pair left concurrent, every φ is invariant across the one I_Φ-trace all linearizations share, so this is exactly $\pi(\mathcal{C}) \models \varphi$ [3,10]. **C4** accepts iff reachability over the finite acyclic data-flow graph finds no unauthorised source→sink path, sound for explicit flows [4]. **C5** accepts iff every (total) Rego guard yields `allow` [2] and the bound of Thm. 3 is $\leq \delta$. As *ok* is their conjunction, $\mathrm{Adm}_G(\mathcal{C}^\star) = 1$. □

Proof (Proof of Proposition 1 (Admissibility-preserving repair)). Every $\mathcal{C} \in [\chi]$ exhibits the witnessed violation of some check C_i, so $C_i(\mathcal{C})$ fails and $\mathrm{Adm}_G(\mathcal{C}) = 0$; refuting $[\chi]$ thus removes only inadmissible plans and U retains all admissible plans. If Propose, complete over U, returns $\perp$, then U contains no admissible plan; as U also contains all admissible plans, none exists. □

Proof (Proof of Theorem 2 (Repair progress and termination)). (i) The loop runs $\leq T$ times and each Verify terminates (each check is a terminating decision procedure). (ii) Witnesses are drawn from finitely many operations, fields, propositions, and bounded-length prefixes, so $|\{[\chi]\}|$ is finite, and an excluded class cannot be re-admitted. If Propose respects R, a returned $\mathcal{C}$ was unrefuted, so refuting its class removes at least $\mathcal{C}$, strictly shrinking U. □

Proof (Proof of Theorem 3 (Distribution-free SLA bound)). The ℓ_i are independent and bounded, so Hoeffding's inequality [5] gives $\mathbb{P}[S - \mu > t] \leq \exp(-2t^2/\sum_i u_i^2)$ for $S = \sum_i \ell_i$. Set $t = B_\ell - \mu$ and solve $\leq \delta$ for the slack condition. A budget violation on availability needs one of m services down, and the union bound gives $\sum_i (1 - a_i)$. □

Proof (Proof of Proposition 2 (Efficient checking, sketch)). C1 is a per-edge subtyping check (linear in k). C2 is k terminating SMT queries over $\mathcal{T}$. C3 tests membership of the single word $\pi(\mathcal{C})$ ($|\pi|{=}k{+}1$) in each $\neg\varphi$ NFA of size $2^{O(|\varphi|)}$, linear in k, and model checking's PSPACE cost is avoided because the LLM supplies the word. C4 is one $O(E)$ reachability pass, and C5 is bounded Rego evaluations plus an $O(m)$ bound. □

2 C3 Ordering: Trace-Closure Soundness

Full LTL$_f$ semantics. Fix atomic propositions AP (action and state markers). A plan induces a finite word $\pi = \pi_0\pi_1 \cdots \pi_{n-1} \in (2^{AP})^*$, where $\pi_i \subseteq AP$ holds the propositions true at step i. Formulas are

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \mathbf{X}\varphi \mid \varphi\,\mathbf{U}\,\varphi \mid \mathbf{Y}\varphi \mid \varphi\,\mathbf{S}\,\varphi, \qquad p \in AP.$$

Satisfaction $\pi, i \models \varphi$ at a position $0 \le i < n$ is defined by

$$\begin{array}{ll}
\pi,i \models p & \text{iff } p \in \pi_i,\\
\pi,i \models \neg\varphi & \text{iff } \pi,i \not\models \varphi,\\
\pi,i \models \varphi_1 \wedge \varphi_2 & \text{iff } \pi,i \models \varphi_1 \text{ and } \pi,i \models \varphi_2,\\
\pi,i \models \mathbf{X}\varphi & \text{iff } i{+}1 < n \text{ and } \pi,i{+}1 \models \varphi,\\
\pi,i \models \varphi_1\,\mathbf{U}\,\varphi_2 & \text{iff } \exists j\ (i \le j < n):\ \pi,j \models \varphi_2 \text{ and } \forall k\ (i \le k < j)\ \pi,k \models \varphi_1,\\
\pi,i \models \mathbf{Y}\varphi & \text{iff } i > 0 \text{ and } \pi,i{-}1 \models \varphi,\\
\pi,i \models \varphi_1\,\mathbf{S}\,\varphi_2 & \text{iff } \exists j\ (0 \le j \le i):\ \pi,j \models \varphi_2 \text{ and } \forall k\ (j < k \le i)\ \pi,k \models \varphi_1.
\end{array}$$

The derived operators are *eventually* $\mathbf{F}\varphi \equiv \mathit{true}\,\mathbf{U}\,\varphi$, *always* $\mathbf{G}\varphi \equiv \neg\mathbf{F}\neg\varphi$, and *once* $\mathbf{O}\varphi \equiv \mathit{true}\,\mathbf{S}\,\varphi$; $\mathbf{X}$ is the *strong* next (false at the last position), and the weak next $\widetilde{\mathbf{X}}\varphi \equiv \neg\mathbf{X}\neg\varphi$ holds vacuously there. A plan satisfies φ iff $\pi, 0 \models \varphi$, and C3 accepts $\mathcal{C}$ iff $\pi(\mathcal{C}), 0 \models \bigwedge_{\varphi\in\Phi}\varphi$. The precedence pattern $\mathbf{G}(b \to \mathbf{O}\,a)$ thus reads "every b is preceded by some a."

Trace-closed safety fragment. C3 scopes Φ to the trace-closed LTL$_f$ *safety* fragment: forbidden finite prefixes, invariant under commuting adjacent independent actions. Each φ names an action set A_φ, and we use the pairwise patterns precedence $\mathbf{G}(b \to \mathbf{O}\,a)$, response, and absence, plus their conjunctions. This fixes the Mazurkiewicz independence relation $I_\Phi = \{(a,b) : \nexists\varphi.\,\{a,b\} \subseteq A_\varphi\}$. C3 *rejects* any plan leaving an I_Φ*-dependent* pair (co-named by some φ, e.g. `charge`,`authenticate`) concurrent, forcing a causal order. Residual concurrency is then only between independent actions, over which every φ in this fragment is invariant, so all linearizations lie in one I_Φ-trace and agree on Φ; hence the single-word test on $\pi(\mathcal{C})$ is sound (vacuous for sequential plans). Formulas outside this fragment, for which pairwise ordering would *not* suffice, are not admitted into Φ.

Worked example. With $\varphi_1{=}\mathbf{G}(\texttt{charge} \to \mathbf{O}\,\texttt{auth})$, $\varphi_2{=}\mathbf{G}(\texttt{charge} \to \mathbf{O}\,\texttt{kyc})$, I_Φ makes $(\texttt{auth},\texttt{kyc})$ *independent* (no φ names both) but each dependent with

`charge`. C3 forces `auth`,`kyc` each before `charge`, leaving their mutual order free; `auth,kyc,charge` and `kyc,auth,charge` agree on Φ, so one word suffices. *Overlapping constraints* only *shrink* I_Φ: adding $\varphi_3{=}\mathbf{G}(\texttt{kyc} \rightarrow \mathbf{O}\,\texttt{auth})$ makes $(\texttt{auth}, \texttt{kyc})$ *leave* I_Φ, since I_Φ is the *intersection* of every constraint's freedom — more policies remove independent pairs, never spuriously add one. Unsoundness could arise only in reverse, if I_Φ declared independent a coupled pair. *Concrete case the fragment prevents:* $\mathbf{F}(\texttt{auth} \wedge \mathbf{X}\,\texttt{kyc})$ is *not* trace-closed (commuting an action between them flips its truth), so two linearizations could disagree; our fragment admits only forbidden-prefix safety patterns, under which every φ is interleaving-invariant, so C3 never asserts more independence than Φ licenses.

3 Relative Completeness

Corollary 1 (Relative completeness). *Call* Propose U-complete *if, given unbounded budget, it eventually returns every member of the unrefuted set U. Then, with $T = \infty$ and a U-complete proposer, VeriSAGE returns a certified plan whenever an admissible plan exists, and* Infeasible *otherwise. Thus VeriSAGE is* sound and complete relative to proposer completeness*: the only way to miss an admissible plan is for the proposer never to propose it.*

Proof. U always contains every admissible plan (Prop. 1). If one exists, a U-complete proposer eventually proposes it and it is certified. Otherwise every proposal is refuted until U empties and Propose returns $\perp$, a sound Infeasible. □

The verifier never errs, so any residual incompleteness is attributable solely to the heuristic proposer, the modular guarantee the design targets.

4 Extended and Correlation-Robust SLA Bounds

Theorem 3 is exact for a *sequential* plan. For a parallel DAG, latency is $\max_{p\in\Pi} \sum_{i\in p} \ell_i$ over source-to-sink paths Π, and C5 certifies the union bound

$$\textstyle\sum_{p\in\Pi} \exp\big(-2(B_\ell - \mu_p)^2 \big/ \sum_{i\in p} u_i^2\big),$$

reducing to Thm. 3 when $|\Pi|{=}1$. Π is small for the near-sequential plans LLMs emit. On wide DAGs C5 sums only the k highest-mean paths (a sound over-approximation) rather than enumerating $|\Pi|$. On failure the check returns a witness and repair seeks a faster path or redundancy. Summing only the top-k paths stays a valid upper bound in the near-sequential regime (omitted paths have strictly lower mean; empirically 100% sound at $\leq$2.1$\times$ true slack, §7).

The Hoeffding tail can be replaced by the variance-aware, still distribution-free *Bernstein* $\exp(-t^2/(2V{+}\frac{2}{3}bt))$ or *Bennett* $\exp(-\frac{V}{b^2}h(\frac{bt}{V}))$ bounds ($V{=}\sum_i \sigma_i^2$ a variance *upper* bound, $b{=}\max_i(u_i{-}\mathbb{E}\ell_i)$, $h(x){=}(1{+}x)\ln(1{+}x){-}x$), needing only a second-moment certificate (from the fitted QWS support,

$\sigma_i^2 \le u_i^2/4$) and certifying the same δ at up to 4.8× less slack under concentration. We evaluate these tighter bounds as an offline analysis (§7, RQ2); the deployed verifier's C5 uses the Hoeffding bound, with Bernstein/Bennett a drop-in refinement (the pointwise minimum) that is never looser. The bound is empirically sound but conservative (§7, RQ2), and distribution-free but not assumption-free: it requires *independent*, bounded-support critical-path latencies.

Correlation-robust fallback (offline analysis). When independence is doubtful (shared-dependency services — 9.5–11.1% of critical-path pairs in our catalogues, §7), a *dependence-free* bound holding for *any* coupling of the marginals can replace Hoeffding: the one-sided Cantelli bound $\mathbb{P}[S-\mu \ge t] \le V^*/(V^*+t^2)$ with worst-case variance $V^*=(\sum_i \sigma_i)^2$ (all correlations +1), or the comonotone Fréchet quantile $\sum_i F_i^{-1}(1-\delta)$. Both cost ∼5× Hoeffding slack at extremal variance but only 1.1–1.6× under concentration (Fréchet a flat ≈1.04×). The Cantelli bound stays 100% sound under injected correlation for *every* coupling up to ρ=1, exactly where independent Hoeffding is *violated* ($\rho \ge 0.9$); the comonotone Fréchet bound stays sound up to high correlation and only marginally exceeds δ at ρ=1 (0.050 vs δ=0.05), i.e. sound for $\rho<1$ (§7, RQ2). A split-conformal certificate [1] would remove the assumption without the worst-case price (future work). Finite-sample error in μ, u_i is absorbed by upper-confidence QWS estimates, so the certified δ already accounts for QoS-estimation error.

Empirical validation (RQ2, measured). The variance-aware, still distribution-free Bernstein/Bennett bounds cut certified slack when latencies concentrate: re-running the 32k calibration at δ=0.05, Bennett needs 22.5% **less** slack under Uniform and 36.1% **less** under Beta(2, 2) (up to 4.8× at ρ=0.01), all 100% sound; only at the two-point law is Hoeffding optimal, so min(Hoeffding, Bennett) is free and never worse. For the correlation-robust fallback, a catalogue scan finds 9.5–11.1% of critical-path pairs plausibly correlated (shared input/producer/backend). Under an injected Gaussian-copula correlation ρ, the independent Hoeffding bound turns **unsound** at $\rho \ge 0.9$ (violation 0.053–0.068>δ=0.05), while the dependence-free Chebyshev bound stays sound for *every* coupling (0.000 up to ρ=1) and the comonotone Fréchet bound stays sound up to high correlation, only marginally exceeding δ at ρ=1 (0.050). The price is ∼5× Hoeffding slack at extremal variance but only 1.1–1.6× under concentration (Fréchet a flat ≈1.04×). For *parallel DAGs* the union-bound conservatism (≈2.6× true slack, 40k MC) is *flat in* $|\Pi|$ (2.56→2.62 for $|\Pi|$=1→27)—the per-path Hoeffding slack, not the path count—and top-mean-path pruning stays 100% sound at 2.1×. Since omitted paths carry their own slack, top-k pruning stays 100% **sound even at fully-tied means** (40k MC, mean-gap 0.6→0); an empirical k-criterion (include every path whose marginal tail contribution exceeds $\epsilon\delta$) adapts k between 6 and 8 as means tie, holding soundness at 100% and conservatism at 2.3–2.8×.

Admission hit on real QWS marginals (Q3, measured). Repeating the correlation analysis with per-service latencies drawn from the *real* QWS marginal

(empirical inverse-CDF, n=2507, support 4989.7 ms) and injected co-location correlation, the picture is milder than the light-tailed worst case: because the distribution-free Hoeffding budget is anchored to the real (large) support, it stays empirically **sound at every** ρ **up to** 1.0 (worst true violation 2.7% $<$ δ=5%) across critical-path lengths m=4–13. The dependence-robust Cantelli/Fréchet fallback — needed in the light-tailed regime where Hoeffding voids at $\rho \geq 0.9$ — costs only $\mathbf{0.8}$–$\mathbf{1.5}\times$ the Hoeffding slack on these real marginals (path-length dependent), so the admission price of dropping independence is modest. Production traces (Alibaba [6], Azure Functions [8]) document the shared-back-end correlation this regime models.

5 Additional Evaluation Detail

Contract-noise robustness (Q4, measured). Soundness is relative to the contract, so we quantify sensitivity to *mis-specified* contracts on 1,500 tasks per point. *Omission* noise (drop a declassifier, add a spurious ordering constraint, over-tag a field, shave the SLA budget) at rate p produces only *conservative* false rejections — 3.1%, 6.8%, 12.1%, 19.0% at p=0.05, 0.1, 0.2, 0.3 — every one of which becomes an ABSTAIN or a repair, never an unsafe plan. *Relaxation* noise that deletes the exact clause guarding an injected violation admits a plan only w.r.t. that *wrong* contract, at 2.6%, 5.3%, 9.6%, 14.6% (below p, since structural C1/C2 violations have no single deletable clause). At p=0 both error modes are 0%: the verifier is exact for the stated contract (the empirical form of Thm. 1). The asymmetry is the design intent — contract error degrades toward abstention, not toward unsafe admission.

Table 1. SOTA agentic baselines on a local (qwen2.5:7b, 3 seeds) and a *frontier* proposer (gpt-4o-mini†, single seed via API); same 100-task suite as Table 3. All baselines are verifier-free during the loop, scored once by the CSTS verifier. On both proposers the verifier-free self-correction/search methods collapse on the hard stratum ($\leq$ 3.3% at 7b, $\leq$ 44% at gpt-4o-mini) while verifier-gated VeriSAGE is best; a frontier proposer narrows the gap to BLIND but not the gap to the verifier-free methods.

	qwen2.5:7b			gpt-4o-mini†		
Method	Overall	Easy	Hard	Overall	Easy	Hard
accept-first (no verifier)	33.3	65.3	1.3	70.0	100.0	40.0
Best-of-N (self-consistency)	33.0	65.3	0.7	70.0	100.0	40.0
Self-Refine (self-critique)	18.3	36.7	0.0	29.0	36.0	22.0
LATS (value-guided search)	33.0	64.0	2.0	69.0	94.0	44.0
CRITIC (schema tool)	51.7	100.0	3.3	70.0	100.0	40.0
blind (verifier-gated)	54.3	100.0	8.7	86.0	100.0	72.0
VeriSAGE (ours)	**55.3**	**100.0**	**10.7**	**87.0**	**100.0**	**74.0**

†hosted frontier model, single seed.

SOTA agentic baselines, full table (RQ4, measured).

Reduced-verifier ablation: are all five classes needed? (measured). On the 446 inadmissible labeled instances, a verifier running only **C1** catches 33.6% of violations; **C1**+**C2** (schema+SMT, the guarantee a typed-tool or consistency checker such as Clover [9] provides) 51.1%; adding explicit-flow (C1–C4) 68.4%; only the full five-class contract reaches 100%. The temporal, information-flow, policy, and SLA classes are precisely what a schema-only admission gate misses — the empirical case for the joint contract.

Verification timing breakdown (RQ3, measured). At k=200 services ($|\Phi|$=10) the per-check structural cost is C1/C2/C3/C4/C5 = 12.2/7.1/17.4/18.7/8.2 μs, and cost grows linearly in $|\Phi|$ at 0.54 μs/formula ($|\Phi| \leq 40$). Under a production full-solver C2 (real QF-LIA+EUF via Z3, 0.30–1.11 ms/service), C2/SMT dominates ($> 99\%$) and non-C2 checks stay 0.28 μs/service, so the end-to-end figure is single-digit-to-tens of ms/admission.

Dependency taint over-approximation and refinement (measured). The static $\mathsf{dep}(s)$ soundly over-approximates the true input→output flow (no false negatives, what C4 needs); the opaque fallback trades precision for soundness. Marking a fraction of operations opaque injects 55–88% spurious dep edges, and on plans with known true flow the induced C4 *false-positive* rate (truly-safe plans wrongly rejected) rises from 16.8% (25% opaque) to 52.7% (fully opaque); a "transparent-preferred" retrieval tie-break cuts this ∼3× (to 5.3–20.3%). **Schema-level taint refinement** does better: keeping an opaque edge $i \to o$ only when the field *types* of i, o are compatible (a real flow needs type-compatible carriage, so this stays sound at 100% recall) raises edge precision 24.5%→72.2% and cuts plan-level C4 false positives 75.1%→36.7%; an optional dynamic-witness layer reaches 100% precision and 8.5% C4-FP (a *may*-analysis, a precision ceiling, not the sound default). Production quantification still needs labelled dependency ground truth (future work).

SLA-bound tightness and acceptance cost (RQ2, measured). The Hoeffding bound is near-tight only against the extremal two-point distribution (empirical/δ reaching 0.046 at δ=0.05, 0.24 at δ=0.30) but very conservative under realistic concentrated latencies (Uniform$[0, u_i]$: empirical violation $< 10^{-3}$ up to δ=0.20, still $1.8{\times}10^{-3}$ at δ=0.30). The acceptance cost (8,000 plans, δ=0.05): under Uniform, 88.5% of plans *truly* meet the SLA yet C5 admits only 59.1%, so **one third (33.3%) of SLA-compliant plans are rejected purely by conservatism**, C5 demanding 2.57× the true headroom (two-point: milder, 73.8% feasible/58.2% admitted/21.2% false-rejected, 1.48×). It also inflates repair: shrinking an over-budget plan to pass C5 takes 2.61 drops (Hoeffding gate) vs. 0.46 (oracle gate; 3,000 tasks, 96.5% incur ≥ 1 extra) — distribution-free soundness at a recall/latency price.

Repair-mechanism ablation, LLM-free (RQ5, measured). Isolating typed compilation from the proposer (deterministic synthetic proposer, feedback the only input, budget-matched T=6, 1–3-defect tasks): typed repair (hard constraint + implied fix) certifies **100**% in 2.2 iterations; enumerate-failing-checks feedback (told *which* checks failed, not the witness) reaches 75.9% at 3.4 iterations and degrades with defect count (92→77→58%); feedback-free resampling 0%. So the typed encoding, not check-enumeration, drives the loop, widening with difficulty.

Multi-defect repair order. On plans simultaneously violating several classes, VeriSAGE repairs in a fixed priority order (structural C1>C2>C3>C4>policy, SLA last, shrinking only distractors), never re-entering a structural class. A sample trace injecting {ordering, policy, functional, SLA} certifies via `C1`→`C2`→`C3`→`policy`→`SLA` in 5 iterations, against 3896 re-entry events for a naive fix-first-firing loop (main paper, RQ5).

A worked repair trace (regulated checkout). The concrete loop behind Fig. 1, as logged. *Proposal 1* `[authenticate, chargeCard, fraudCheck, ship]` writing `card.pan`→`log`: VERIFY short-circuits at C3 with witness "`chargeCard` precedes required `fraudCheck`"; COMPILEREPAIR adds the hard constraint `fraudCheck`≺`chargeCard` and the prompt "run the compliance check before charging." *Proposal 2* `[authenticate, fraudCheck, chargeCard, ship]`: C3 now holds but C4 fails — "`card.pan` reaches public sink `log` un-declassified"; COMPILEREPAIR adds "no `pii` field may reach `log`" and the prompt "tokenise the card before logging." *Proposal 3* `[authenticate, fraudCheck, tokenize, chargeCard, ship]` routing `card.tok` (public) to the sink: all of C1–C5 pass and the plan is **certified** ($\mathcal{C}^\star$). Two typed counterexamples, two hard constraints, one certified plan; each repair fixes the witnessed class without disturbing the others.

Remaining full-scale comparison (design, not measured). The remaining comparison pits VeriSAGE against frontier production agents (Direct-LLM GPT-4o/Claude, ReAct, Reflexion, ToolBench, a graph-based LLM composer, QoS-MILP, and a genetic variant) over the full 340-task VeriSAGE-Bench on nine metrics. Any significance tests (per-task McNemar/Wilcoxon, Holm-Bonferroni) will be computed *only* on measured paired outcomes; the harness is designed and released, but reporting the comparison needs a frontier-model budget we lack, so it is deferred and no projected numbers appear in the paper.

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